Thesis Outline

Please note, this is a suggested structure to give you an idea of what chapters to include and what the chapters should contain. As each research project is different, your thesis may contain additional chapters/ sections.

Abstract:

This is a bird's eye view of your entire thesis. It is a synopsis of your research from start to finish, outlining your main findings and academic contributions. It should be less than 300 words. (You shouldn't have any references in the abstract)

Introduction:

Include background, problem definition (what is the problem? and why it's a problem? What impact does it have?)
(Include any relevant stats/ figures to support your research)

- **Aims:** What you are trying to achieve with your research
- **Objectives:** How you will achieve those aims
- **Hypothesis**: the question or statement your research is trying to prove (or disprove)
- **Research questions**: the questions you asked (or should have asked) before you started your research. These questions will tie in with your conclusions (i.e., have they been answered and if not, why)
- **Contributions:** What are the contributions you are making to your area of research?
- **Chapter overview:** Brief overview of the rest of the dissertation. E.g. "chapter 2 discusses the research related to my chosen topic..." etc.
- **Terminology:** Any specific terminology that may need explaining.

Chapter 2 (Literature Review or Related Research)

This chapter is important because it establishes that you have looked into anything related to your own research (Answers the question "what's been done by others in my area of study?"). Note that the literature review should not contain explanations of technologies or theoretical "text-book" information. If you must explain background technologies or underpinning theories, you could write them into a separate chapter.

This chapter will reveal any shortfalls or gaps in research (which you can fill with your research), so justifies why you are undertaking this project.

Citations and referencing are very important throughout the thesis, but even more so in this chapter. Research reviewed should be mainly published, peer-reviewed work, i.e., research and journal papers, cited white papers, books, papers from trusted Websites (e.g., SANS) etc. Use the college library databases (e.g., IEEE, ACM, Springer and Science Direct) for full paper access. You can access the databases on or off campus.

Note: A review is **not just a summary**. You should be able to critically assess the research also. You should try to identify the limitations of each approach. This helps support the need for your research.

You should include a short summary at the end of your review, including your main findings from the related works and where your research fits into the overall landscape (and how you are improving the problem area).

Chapter 3: Methodology

This chapter should describe the framework for your research methods, i.e. how you went about researching your project. It should describe how you carried out your research, the tools and techniques employed, where your data (if applicable) comes from, what sort of data analysis techniques you used, and any measurements or metrics used to support your work. Note that you not only need to explain how research was conducted, but also why you used these methods and how they served your purposes. You should link your methodology back to your literature review to support your reasoning for using certain methods.

If your project involves some software/ tool development, you should include a description of how this was designed/built and what purpose it serves (how does it help in your research?). You should include an explanation of the functionality and a brief walkthrough of how it works (avoid too much "step-by-step" explanations, which will make it look like a tutorial).

Chapter 5: Experimental Analysis

Detail how you performed your experiments. Again, this should not read like a "how to" manual or tutorial. You can talk briefly about the set up, but don't go into step by step detail of how you installed each tool or software. Likewise, you shouldn't give a step by step account of how you used any tools; summary information is enough e.g.:

"*By correlating the data extracted from process monitor and regshot, it was evident that the malware set up a persistence mechanism in the startup folder. This was found to be a file called evil.exe, located in C:\Users\Fred\Appdata\Local\Temp folder*"

Concentrate on the experiment itself, what you were trying to prove and the steps involved in performing the experiment. This chapter may include a comparison with state-of-the-art tools or a benchmark dataset to demonstrate the relevancy and capabilities of your research methods.

Chapter 6: Results and Interpretation

You should now have some data or results back from the experiments. This chapter attempts to make sense of the results of your analysis. For example, if you're comparing data taken from a manual analysis with that taken from an automated tool, you would discuss the results from each method and what the advantages and disadvantages of each method were. If you have a lot of stats data, summarise it using graphs.

Chapter 7: Conclusions:

Answer questions like:

- What has been achieved by your research?
- Have you answered your research questions? If not, why?
- Have you proved or disproved your hypothesis?
- Has your research contributed to your chosen field and how?

Chapter 8: Further work

There will be limitations to your research, since you only have a short amount of time, so you should identify these and give a short explanation of how you could address these to extend or improve your work. This chapter can be included in the conclusions if it is short.

References: Choose a particular style of referencing and stick to that (e.g., Harvard or IEEE).

Appendices: Appendices are used for information, such as code, figures, data and tables that may not belong in the main body of the text (e.g., if they are supplementary or overly long).

NOTE: If your research involves a lot of coding, don’t add it in as an appendix. You can put the code onto a repo such as GitHub and add the link in your thesis or upload it along with your thesis.